

Using Logistic Regression and XGBoost for Bot Detection

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Abstract

The activity of bots on social media platforms is increasingly becoming a problem and platform providers are seeking ways to prevent bots from infiltrating their systems. The following project evaluates two simple machine learning algorithms on their ability to discern between bots and users based on their usage data. The algorithms applied are logistic regression (LR), which was able to reach an accuracy of 97.28%, and XGBoost (XGB), which reached 97.70% accuracy.

1 Data and Preprocessing

1.1 Dataset Description

The "bots_vs_users.csv" dataset contains information on 5874 social media accounts, exactly half of which are bots. Every account has 59 features, which include 14 numerical ones like their `average_views` and 45 categorical ones like the city they live in. 41 of the categorical features are binary, like the variable `has_profile_picture`. Most of the features include a large number of missing or unknown values. 4483 accounts don't contain any numerical information, while for the individual categorical features, between 5006 and 24 of the accounts carry the label "unknown". Only the target variable "target", which indicates if the account is a bot or not, has no missing values.

1.2 Data split

Before any preprocessing was applied, the data was split into a training and a test set using a stratified 80/20 split. Model hyperparameters were validated based on a k-fold cross-validation on the training set. To ensure reproducibility, this and every following step that involves randomness was seeded with the seed 42.

1.3 Preprocessing and Feature Engineering

The following section lists all transformations in the order they were applied. Each transformation that requires learning distributional features was fit only on the training set to prevent data leakage.

1. **Minimal profile flag:** A binary categorical feature was added, distinguishing between accounts with and without initial numeric feature values.
2. **Subscribers count feature:** The subscribers count feature was transformed from a categorical feature with 1597 categories into a numeric feature.
3. **Data scaling:** To ensure equal contribution of features and faster convergence for LR, the data was scaled and skew was removed using the Yeo-Johnson method of `sklearn`'s `PowerTransform` class. (Yeo & Johnson, 2000)
4. **Median imputation:** To remove NaN values, the `SimpleImputer` class of the `sklearn` library was used to replace all missing values for numerical features with the median of their feature.
5. **Categorical encoding:** Categorical features with exactly two categories were binary encoded, mapping each category to 0 or 1. The remaining categories were one-hot encoded, while assigning all categories with fewer than 30 members to a "other" bin. This reduced the number of city categories from 362 to only 6, preventing a drastic dimensionality increase of the data. For all categorical features, the `unknown` category was treated just as any other category.

2 Models

2.1 Baseline

For a baseline measurement, `sklearn`'s `DummyClassifier` was used with the "stratified" strategy. This strategy randomly predicts

labels based on the frequency at which they occur in the dataset. It achieved an accuracy of 0.5 and an F1-score of 0.49 on the bot class.

2.2 Logistic Regressor

LR is one of the simplest machine learning algorithms. Despite its simplicity, it performs surprisingly well on linearly separable classification tasks. A major advantage of LR is that its coefficients directly reflect feature importance for the classification task and can thus easily be interpreted. Additionally, it is very computationally efficient when dealing with large feature spaces. This makes it a good choice for local hyperparameter tuning and evaluation on the current dataset.

2.3 XGBoost

Extreme Gradient Boosting, known as XGBoost (XGB), is a high-performance implementation of gradient-boosted decision trees, designed for speed and efficiency. (Chen & Guestrin, 2016) As other decision tree-derived algorithms, it is particularly well-suited for structured datasets with a mix of categorical and numerical features, as in the current case.

2.4 Model Comparison

The most striking difference between LR and XGB is their ability to handle nonlinear data. While XGB can pick up on complex nonlinear relationships, LR doesn't have this ability. Next to this difference, the two algorithms are similar in that they both are very computationally efficient and use regularization to decrease overfitting, which is essential when working with high-dimensional data.

3 Model Training

3.1 Logistic Regression

A critical hyperparameter in LR is C , which controls the strength of regularization. Lower C values impose stronger regularization, shrinking the model's coefficients toward zero (in the case of the default L2 penalty), and thereby reducing the risk of overfitting.

A grid search was performed using `GridSearchCV` from `scikit-learn` to select the optimal regularization parameter C , testing the values 0.01, 0.1, 0.25, 0.5, 1, 2.5, 5, 10, 20, 40, and 80. $C = 2.5$ was identified as the best value, and the final LR model was trained with this hyperparameter on the whole training set.

As visible in Figure A.1 in the appendix, the average F1 score across five folds peaks around the value of 2.5, confirming the value arrived at with gridsearch. Nevertheless, the error bars indicate that the effect is extremely small. The difference between the worst and the best mean F1 score is only around 0.004 points, indicating that regularization strength doesn't have a strong impact on performance in this dataset.

3.2 XGBoost

Gridsearch proved unsuitable for optimizing the hyperparameters of XGB because the number of possible combinations is too large for local computation. The open-source hyperparameter optimizing framework Optuna (Akiba et al., 2019) was used instead as it automates the efficient search through a large space of possible hyperparameter combinations. Optuna uses advanced strategies like Bayesian optimization to intelligently explore the search space, thereby significantly reducing the number of tests that need to be done to arrive at a good result.

To ensure reliable results, every hyperparameter combination was scored on its average F1 score over 5 independently seeded 3-fold cross-validations. Optuna conducted a total of 100 trials with different combinations of hyperparameters. The best performing configuration achieved an average F1 score of 0.9781. Search ranges for hyperparameters and the resulting best combination can be found in Table 1.

Table 1: Best hyperparameter configuration for XGBoost and corresponding search ranges

Hyperparameter	Best Value	Search Range
<code>n_estimators</code>	281	[50, 500]
<code>learning_rate</code>	0.2003	[0.01, 0.3]
<code>max_depth</code>	3	[3, 8]
<code>min_child_weight</code>	1	[1, 5]
<code>subsample</code>	0.9284	[0.6, 1.0]
<code>colsample_bytree</code>	0.6527	[0.6, 1.0]
<code>gamma</code>	1.0618	[0, 5]

Like previously with the logistic regressor, variance between the best and worst set of hyperparameters was only minimal, this time spanning a little more than 0.005% in average F1 performance. Due to the fact that different hyperparameters are interacting, their individual values become less meaningful. Figure A.2 attempts to visualize their interaction by plotting how learning rate and gamma influence the F1 score. It reveals that a learning rate

around 0.2 results in slightly higher F1 scores, while higher gamma values tend to slightly decrease prediction performance.

The final model used the best parameters listed in Table 1 and was re-trained on the entire training set.

4 Results

4.1 Logistic regression

On the hold-out test set, LR reached an F1 score of 0.9724, accuracy of 0.9728, precision of 0.9843, recall of 0.9608, and an area under the curve (AUC) for the receiver operating characteristic (ROC) of 0.9951. Table A.1 displays the confusion matrix of the predictions, revealing that only 23 out of 587 bots got misclassified as users. A ROC curve and a precision-recall curve can be found in figure A.4 and figure A.3 of the appendix, respectively.

4.2 XGBoost

XGB achieved an F1 score of 0.9768, accuracy of 0.9770, precision of 0.9861, recall of 0.9676, and ROC-AUC of 0.9962. Those metrics are reflected in its confusion matrix (Table A.2) that shows only 19 misclassified bots. Like previously with LR, the ROC curve (Figure A.6) and a precision-recall curve (Figure A.5) are very close to the optimum.

5 Discussion

5.1 Model Performance Comparison

While XGB performs slightly better on all metrics, this difference is small. With only 19 instead of 23 misclassified bots and only 8 instead of 9 misclassified users, XGB’s accuracy is 0.0042 points higher than that of LR. This is also reflected in the F1 score, where the difference increases slightly to 0.0044 points. Those differences are incredibly small and clearly show that XGB’s ability to find nonlinear relationships in data is not majorly critical for the given dataset.

5.2 Generalizability

Both models seem to generalize extremely well and don’t show signs of overfitting. Comparing the F1 scores during hyperparameter tuning and testing shows only minimal differences of 0.0013 points in the F1 score of XGB and 0.0022 for LR. These differences are extremely small, especially when considering that

hyperparameters were specifically chosen to fit the training set as well as possible.

5.3 Feature Analysis

The `Explainer` class of the Python `shap` library was used to extract SHAP values for the different features. Figures A.7, A.8, A.9, and A.10 visualize those, showing a large overlap with the six most important features of LR all being in the top eight most important features of XGB. Most notably, the one-hot encoding of the unknown class in the `has_status` feature is the most important feature for both algorithms. Looking at the gender variable reveals that the error rate decreases with increasing frequency per bin in both algorithms. (Table A.3) The positive class has the most examples and the lowest error rate, while the unknown class has the fewest examples and the highest error rate, which is caused by only one mistake.

6 Reflection

Through consistent seeding, the results of this report can easily be reproduced. A potential improvement would be to check different penalties and solvers for LR as further hyperparameters. Due to the large amount of effort that went into optimizing XGB hyperparameters, this was omitted, and default parameters, which tend to be optimal in most cases, were used.

A possible problem for reproducibility is that different seeds are likely to result in different combinations of hyperparameters. This happens because the variance created by hyperparameter choice is far lower than the variance created by different splits of the data (see Figure A.1). This fact decreases the reliability of the "best hyperparameters" arrived at in this report. During tuning of XGB hyperparameters, variability was decreased by systematically varying seeds and testing each hyperparameter combination on five different splits. This approach decreased the amount of noise in the results, but can not fully eliminate it. Testing for longer and including even more different splits per combination would improve reliability further. Fortunately, the differences in performance are minimal for most hyperparameter combinations, ensuring that the current result will not be far from the optimal one.

References

- Akiba, T., Sano, S., Yanase, T., Ohta, T., & Koyama, M. (2019). Optuna: A Next-generation Hyperparameter Optimization Framework. *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery & Data Mining*, 2623–2631. <https://doi.org/10.1145/3292500.3330701>
- Chen, T., & Guestrin, C. (2016). XGBoost: A Scalable Tree Boosting System. *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, 785–794. <https://doi.org/10.1145/2939672.2939785>
- Yeo, I.-K., & Johnson, R. A. (2000). A new family of power transformations to improve normality or symmetry. *Biometrika*, 87(4), 954–959. <https://doi.org/10.1093/biomet/87.4.954>

A Tables

Table A.1: Confusion matrix for logistic regression predictions on the test set

	Predicted Users	Predicted Bots
Users	579	9
Bots	23	564

Table A.2: Confusion matrix for XGBoost predictions on the test set

	Predicted Users	Predicted Bots
Users	580	8
Bots	19	568

Table A.3: Error counts, total samples, and error rates by gender

gender	LR Errors	XGB Errors	Total Members	LR Error Rate	XGB Error Rate
1.0	19	17	888	0.021	0.019
2.0	12	9	278	0.043	0.032
other	1	1	9	0.111	0.111

B Figures

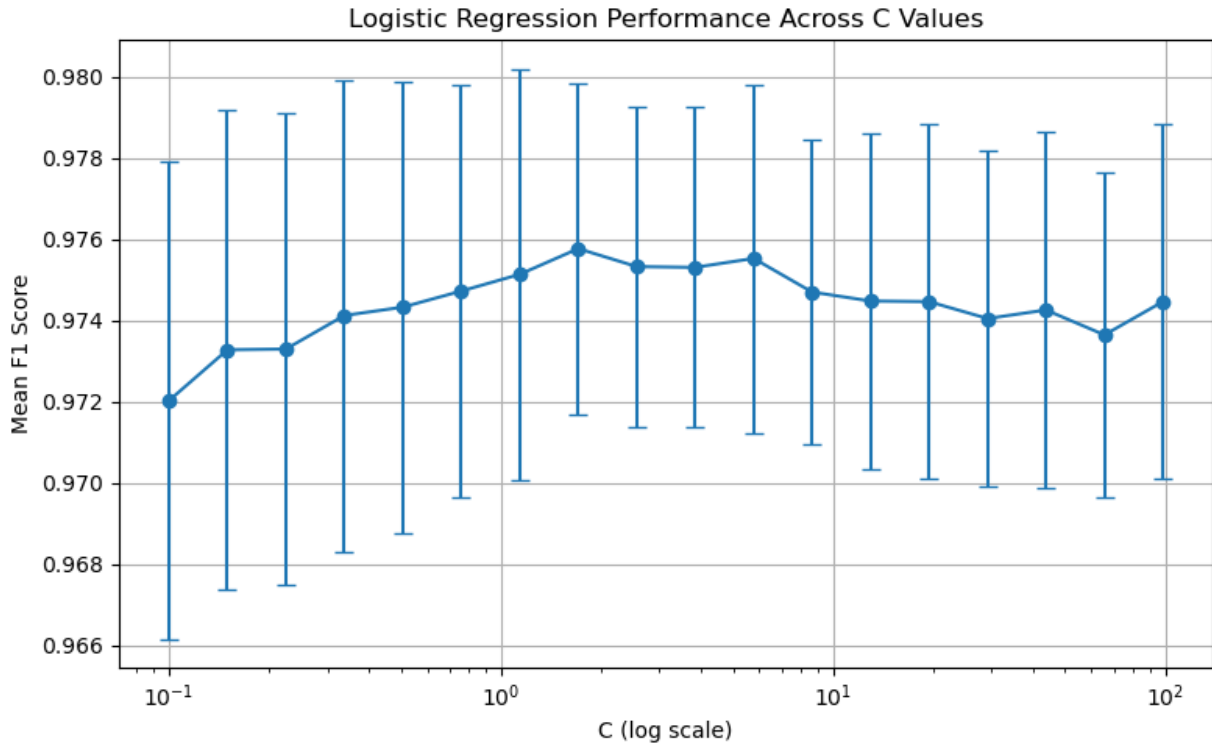


Figure A.1: Mean F1 score of logistic regression across different regularization strengths (C), evaluated using 5-fold stratified cross-validation. Error bars represent the standard deviation across folds. A log scale is used on the x-axis to capture the exponential range of tested C values. While a peak around 2.5 is visible, the error bars suggest that the effect of C is only minimal.

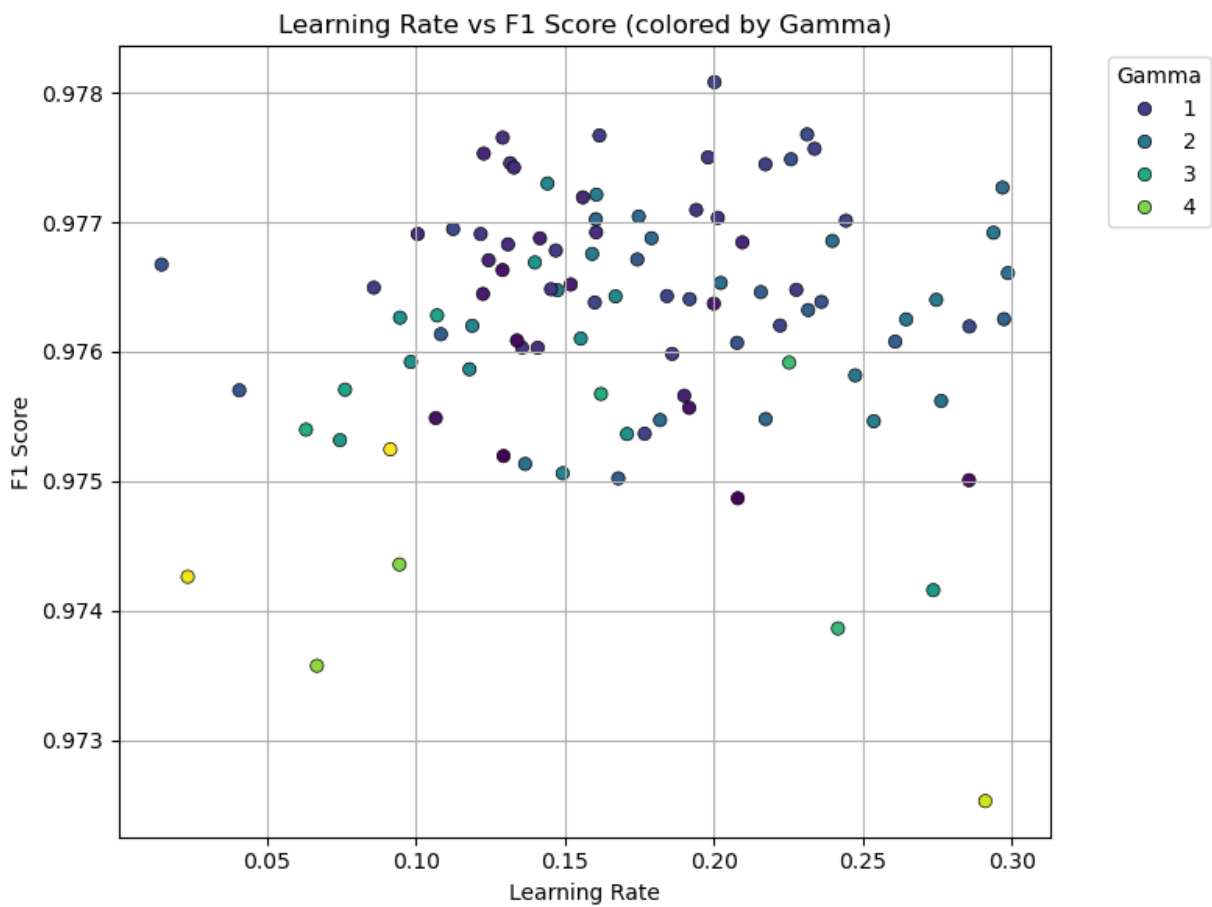


Figure A.2: Scatterplot of the effect of the XGBoost hyperparameters `learning_rate` and `gamma` on average F1 score.

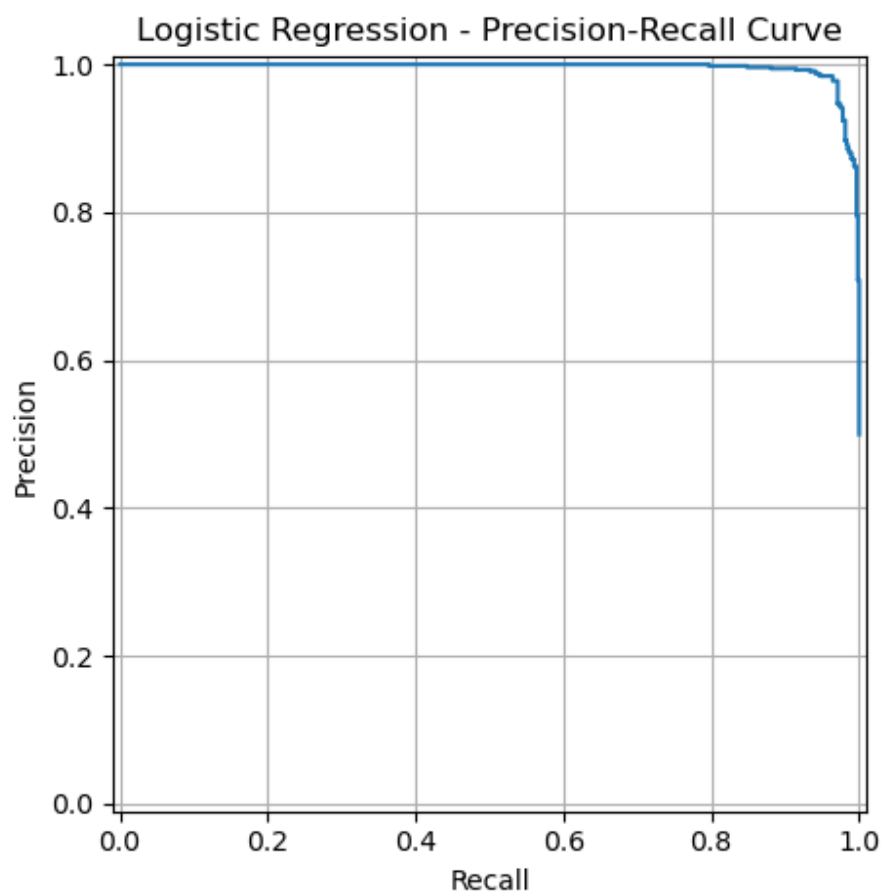


Figure A.3: Precision recall curve of the logistic regressor on the hold-out test set.

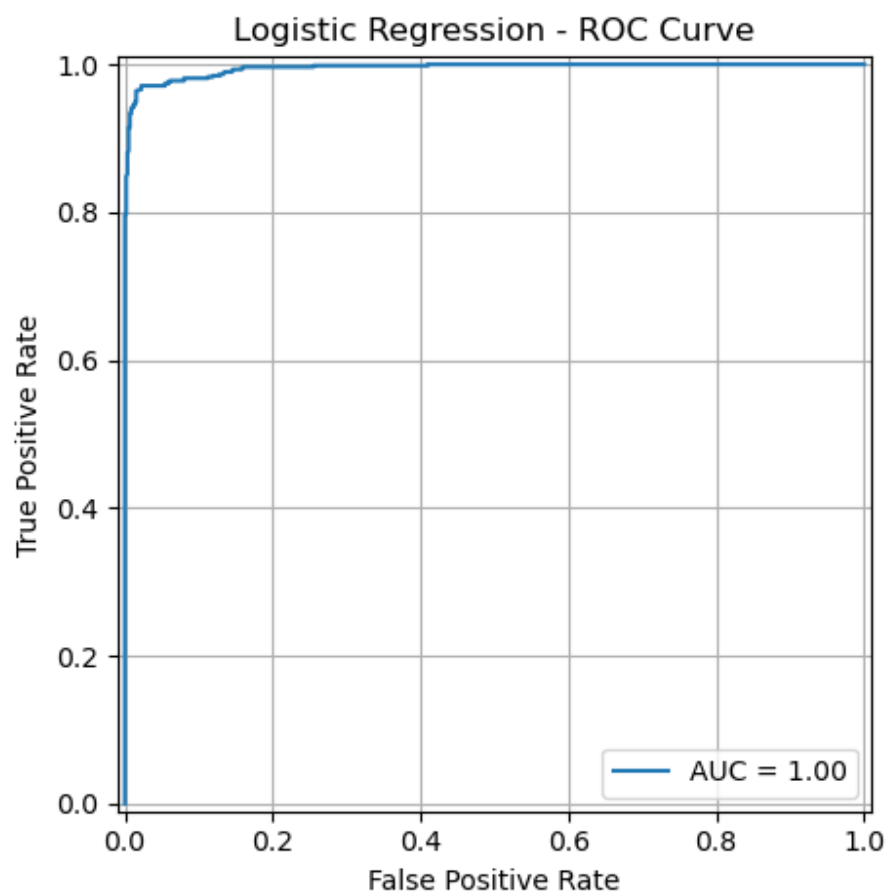


Figure A.4: Receiver operating characteristic curve of the logistic regressor on the hold-out test set.

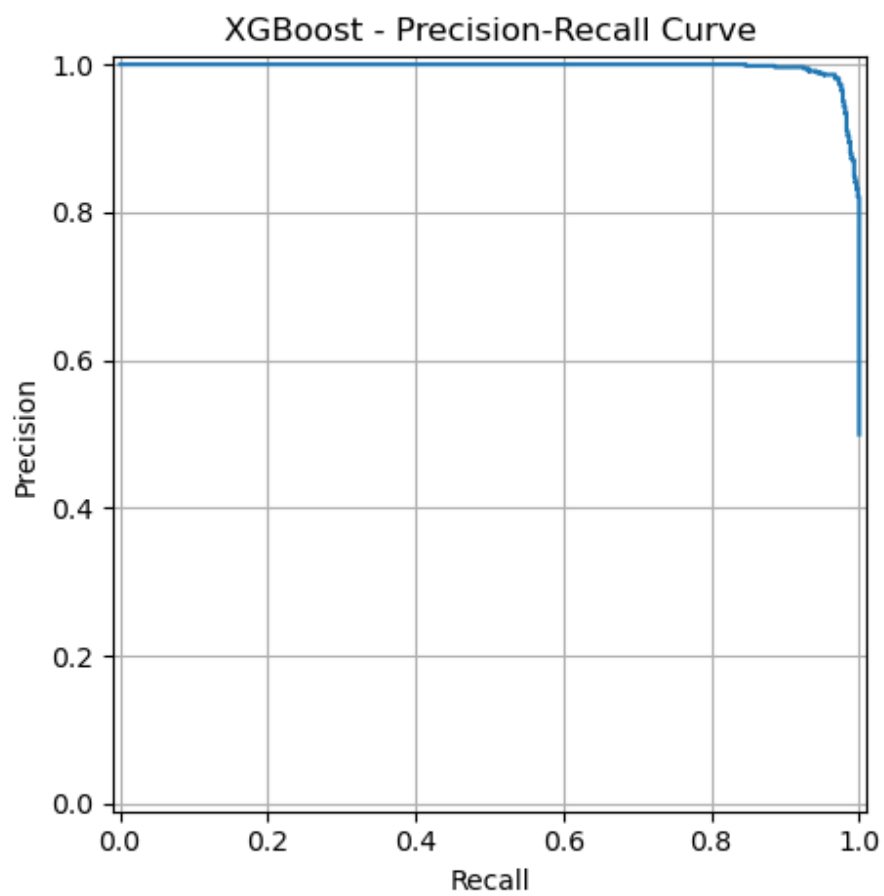


Figure A.5: Precision recall curve of XGBoost on the hold-out test set.

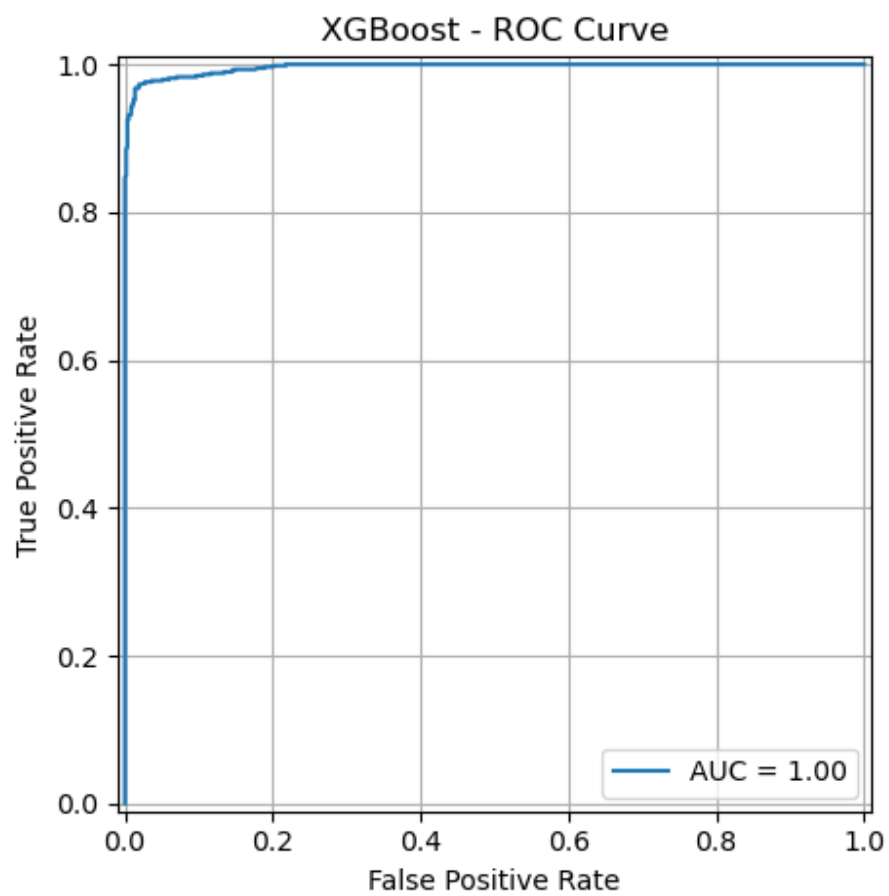


Figure A.6: Receiver operating characteristic curve of XGBoost on the hold-out test set.

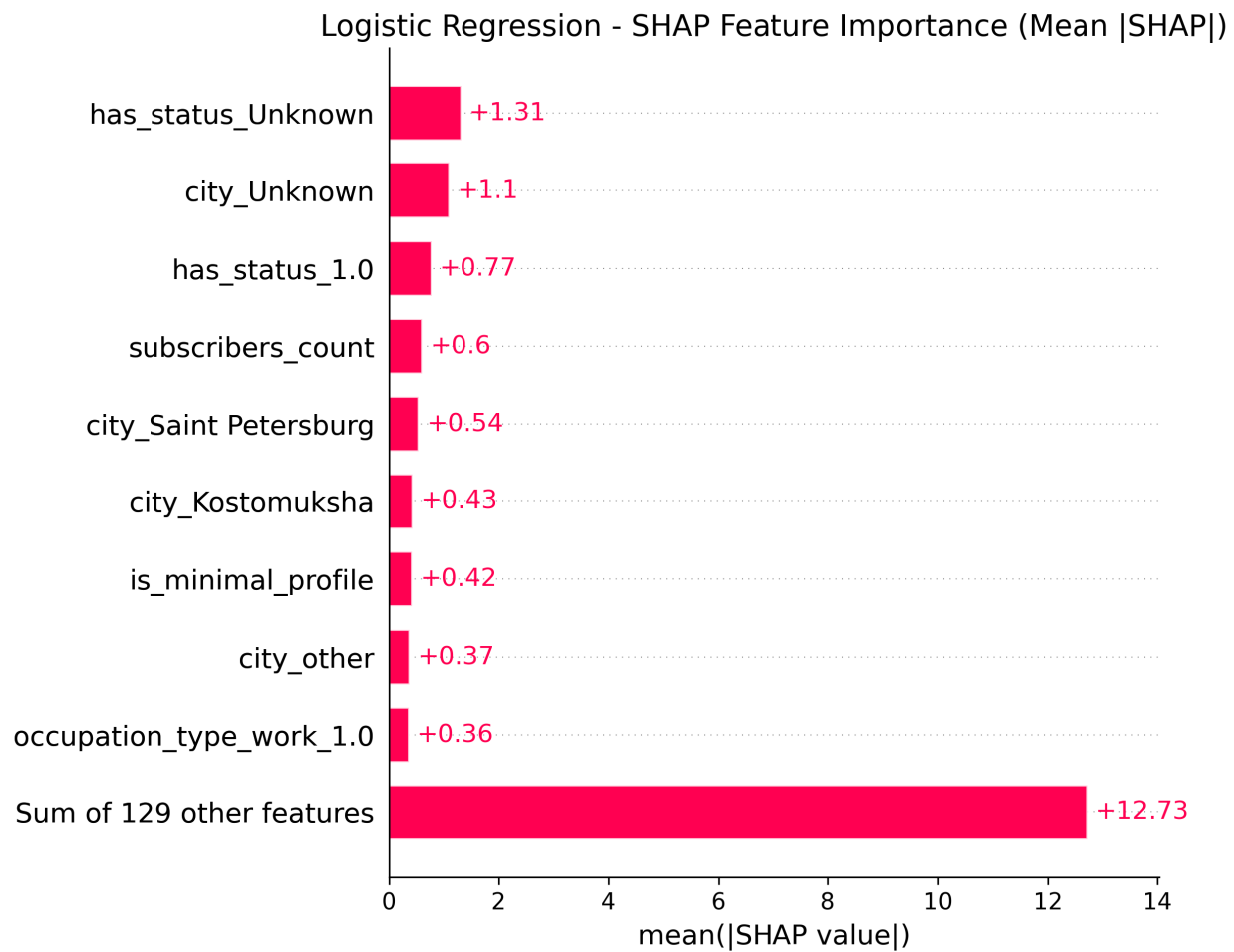


Figure A.7: Bar graph, comparing SHAP feature importance of the 9 most important features in the logistic regressor with the summed importance of the 129 remaining features.

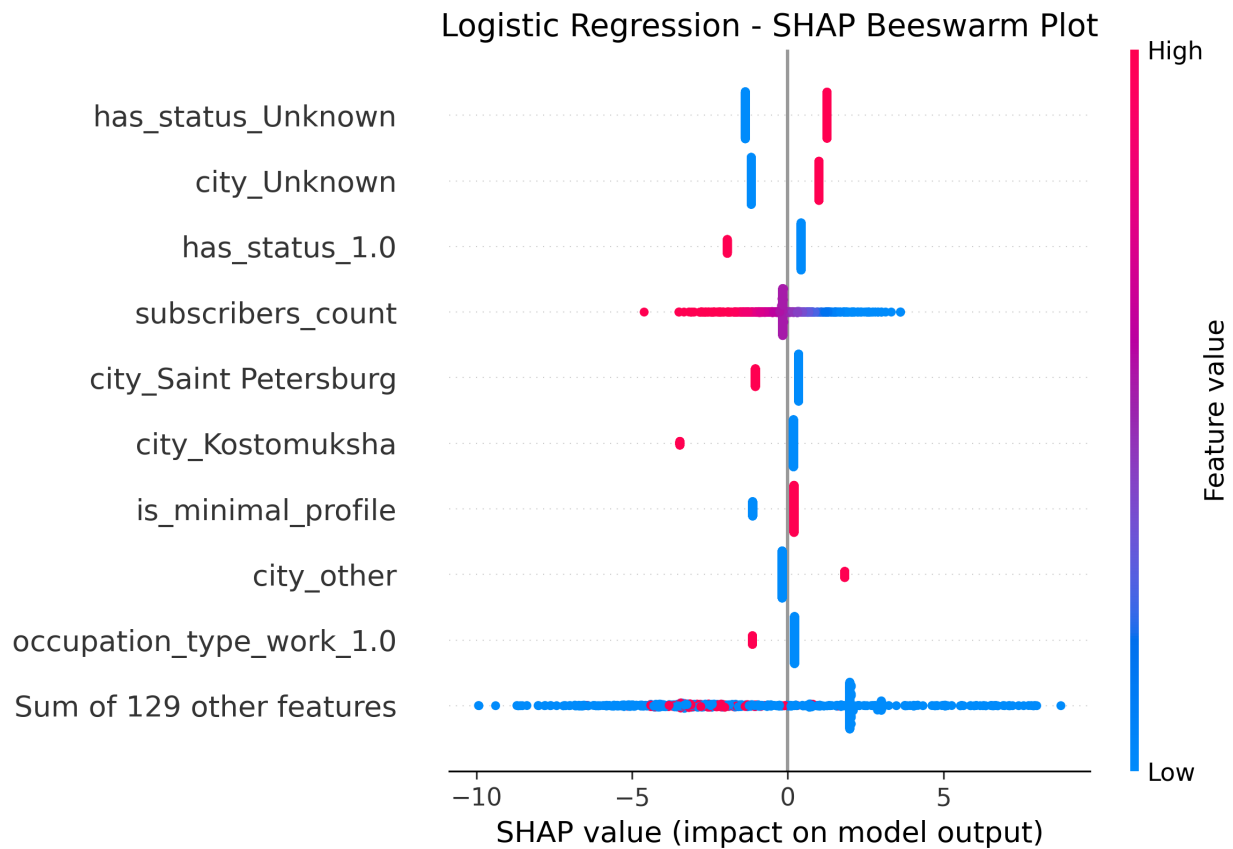


Figure A.8: SHAP beeswarm plot showing the global feature importance for the logistic regressor. Each dot represents a SHAP value for a single prediction and feature. The horizontal position indicates the impact on the model output (positive values push the prediction toward the bot class), while the color reflects the original feature value (red = high, blue = low). Features are sorted top-to-bottom by overall importance.

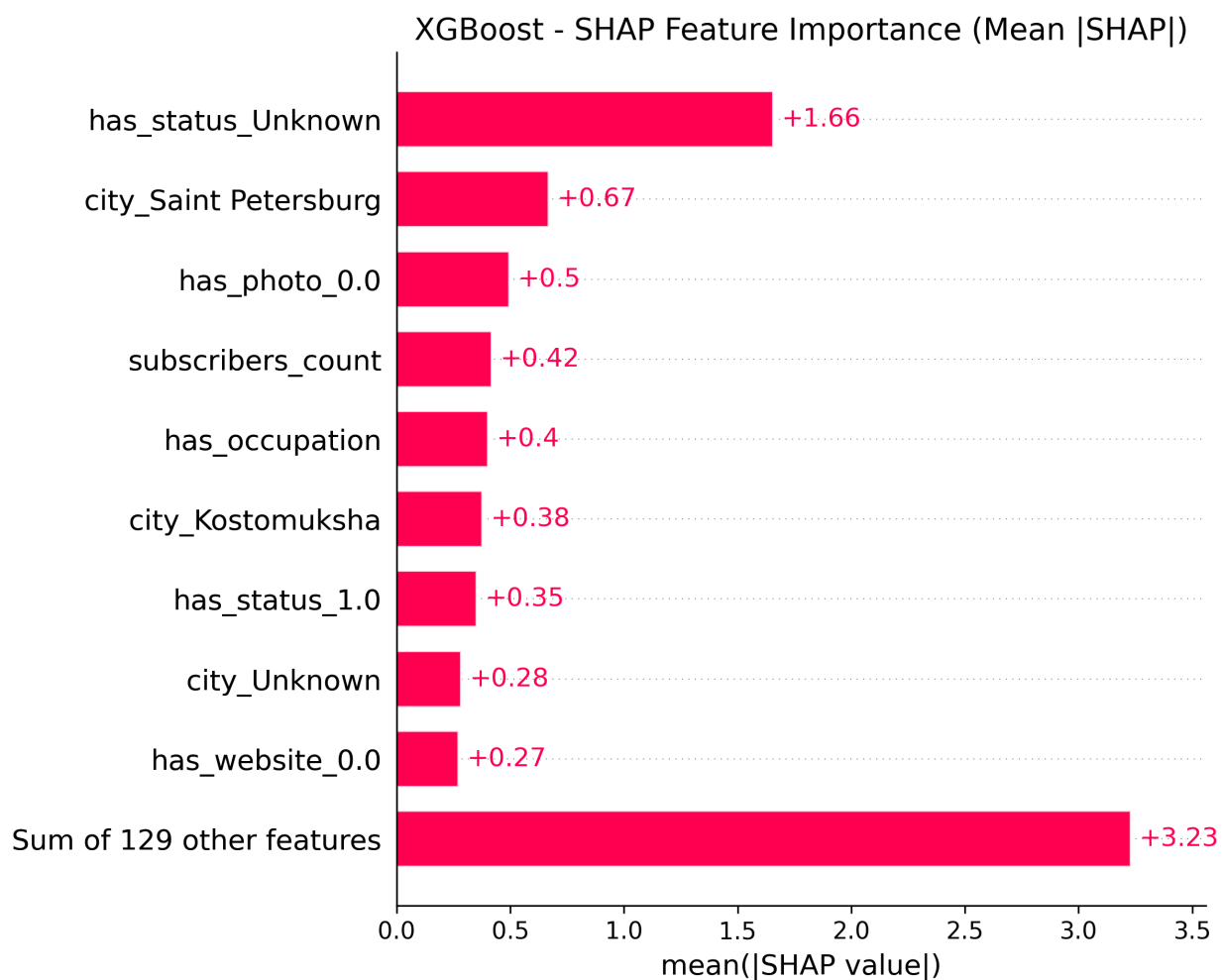


Figure A.9: Bar graph, comparing SHAP feature importance of the 9 most important features of the trained XGBoost model with the summed importance of the 129 remaining features.

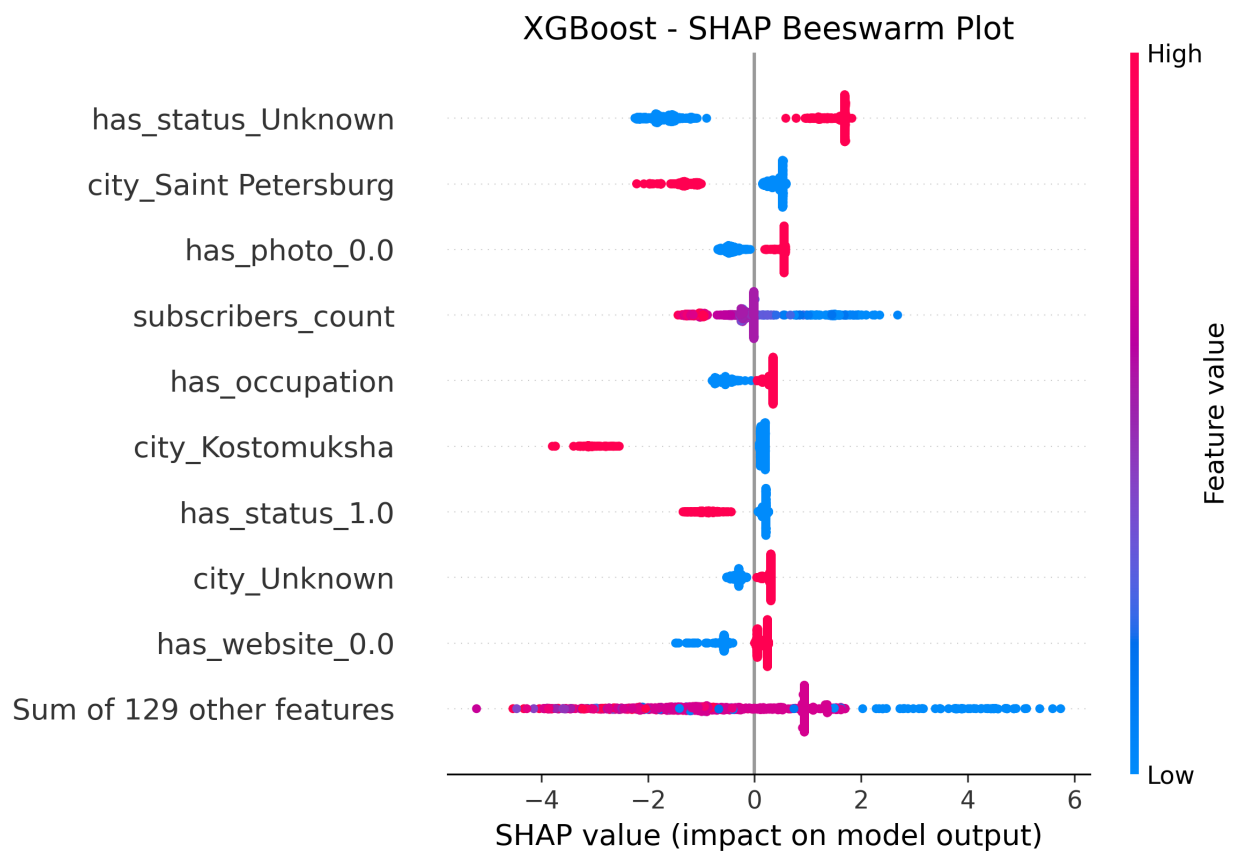


Figure A.10: SHAP beeswarm plot showing the global feature importance of the trained logistic regression model. Each dot represents a SHAP value for a single prediction and feature. The horizontal position indicates the impact on the model output (positive values push the prediction toward the bot class), while the color reflects the original feature value (red = high, blue = low). Features are sorted top-to-bottom by overall importance.